Unit 4 · Outcome 2 · Energy Sources

Where does the energy actually go?

KK05 VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary — clean explanations, comparison tables, visual aids, exam traps, command words and worked answers.

THE BIG IDEA

Energy is never used up — it's transformed, and degraded

A power station doesn't make energy. It converts energy that's already stored in something (chemical bonds in coal, water held high behind a dam, moving air) into the form we want: electrical energy.

The First Law of Thermodynamics says energy can't be created or destroyed — only changed from one form to another. So if you put 100 units of energy in, 100 units come out. The catch is which forms they come out as. We want electricity. What we usually also get is a pile of low-grade heat spilling into the air and water — energy that's still real, but useless for running a fridge.

An exam conversion chain is built from these forms. Learn the colours — we use them all through this module.

Energy stored in bonds — coal, gas, biofuel, food, batteries. Released by combustion or reaction.

Stored by position — water held high behind a dam. Gravitational PE = mgh.

Energy of motion — falling water, moving air, spinning turbine blades. KE = ½ mv ^{2}.

The total working energy of moving machinery (kinetic + potential of the moving parts) — e.g. a turbine shaft.

Heat — random motion of particles. The form most energy wastes into. Hot steam, hot gases.

The useful output we're chasing — energy carried by moving charge through the grid.

> ◈ **ANALOGY**
>
> Analogy — the leaky-bucket relay Picture a fire brigade passing buckets of water hand to hand. Every single handover spills a little. With one handover you lose a splash; with four handovers you lose four splashes. The water that reaches the fire is the useful output; the puddles on the ground are the wasted heat. The more conversions in the chain, the more total spillage — that's exactly why multi-step generation is less efficient.

THE ONE FORMULA

Efficiency = useful out ÷ total in

This is the single equation behind every calculation in this dot point. Memorise it, and memorise that the answer can never exceed 100%.

> • **NOTE**
>
> η ("eta") = efficiency. Works with energy (J, kJ, MJ, GJ) or power (W, kW, MW) — as long as both top and bottom use the same units.

A question might give you energy (joules — a total amount) or power (watts — a rate, joules per second). The formula is identical. Just keep the units consistent.

By the First Law, everything in must come out. So the energy that isn't useful is the rest — almost always lost as thermal energy (heat) to the surroundings.

The generator at Yarrabin's hydro plant receives 500 MJ of kinetic energy from the spinning turbine and delivers 475 MJ of electrical energy. What is its efficiency?

Wasted: 500 − 475 = 25 MJ lost as heat in the windings and bearings. Notice the answer is below 100% — it always is.

INTERACTIVE · SCENE 02

The conversion-chain lab

Each box shows the energy surviving after that conversion. The overall efficiency is the running product — multiply the step efficiencies, never add them.

THE EXAM-CRITICAL SKILL

Multi-step: why the losses compound

When energy passes through several conversions, you multiply the step efficiencies (as decimals) — you never add or average them.

• **NOTE**

Then × 100 to express as a percentage. The overall figure is always lower than the worst single step.

◈ **ANALOGY**

Analogy — three middlemen each take a cut Say your pocket money passes through three people, and each keeps a slice. The first lets 90% through, the second 50%, the third 95%. You don't subtract their cuts — you keep 0.90 × 0.50 × 0.95 = 0.4275, i.e. about 43%. More middlemen (more conversion steps) means less reaches you, fast.

Coal makes electricity through a four-form chain. Each conversion has its own efficiency:

So roughly two-thirds of the coal's energy is thrown away — mostly as waste heat at the steam-turbine step. Real Victorian brown coal is even lower (~25–30%) because wet brown coal carries a lot of water that must be boiled off first.

The killer step above is thermal → kinetic (only 42%). That's the heat-engine bottleneck — turning heat into motion is fundamentally limited by the Second Law of Thermodynamics (the next dot point, KK06). Any source that burns something — coal, gas, biomass, even nuclear — is stuck with this step.

Hydro and wind don't burn anything. Their energy is already mechanical, so they skip the thermal stage entirely:

Source	Conversion chain	Typical overall efficiency
Hydro	Potential → Kinetic → Mech → Elec	~90%
Wind	Kinetic → Mech → Elec	~40%*
Gas (CCGT)	Chem → Thermal → Kinetic → Elec	~55%
Black coal	Chem → Thermal → Kinetic → Elec	~38%
Brown coal	Chem → Thermal → Kinetic → Elec	~28%
Solar PV	Light → Elec (1 step)	~20%

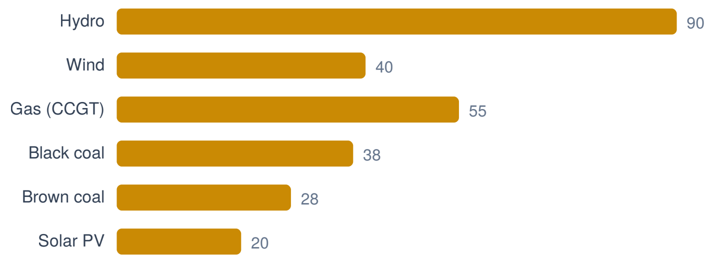

Figure · Source compared at a glance.

*Wind is one step short but limited by the Betz limit — a turbine can extract at most 59.3% of the wind's kinetic energy. Solar PV has just one conversion yet sits low because of the physics of the photovoltaic cell, not because of a long chain. Lesson: fewer steps usually helps, but each step's own ceiling matters too.

INTERACTIVE · SCENE 03

Build-your-own Sankey diagram

A Sankey diagram draws energy as a flow whose width is proportional to the amount. The useful stream stays on the main path; wasted heat branches off. Drag the three step efficiencies and watch the river narrow.

Try dragging step 2 (the turbine) up to 100% — see how much the whole river fattens. That single step is the heat-engine bottleneck doing all the damage. Examiners love asking you to read useful vs wasted straight off a Sankey.

WORKED EXAMPLES

See the marks being earned

Three classic question types. Open each, cover the working, and try it first.

Yarrabin's gas plant converts chemical energy → thermal (88%) → kinetic (60%) → electrical (96%). Calculate the overall efficiency. (2 marks)

Marking: 1 mark for multiplying (not adding) the three decimals; 1 mark for the correct answer with a % sign. Round sensibly and keep the unit.

The solar farm receives 1500 MJ of light energy in an hour and operates at 22% efficiency. Determine the electrical energy generated and the energy wasted. (3 marks)

Marking: rearranging the formula correctly (1), the 330 MJ output (1), the 1170 MJ waste by subtraction (1). Note the waste is not destroyed — First Law — it leaves as heat.

A biomass plant has an overall efficiency of 30%. Its boiler (chem→thermal) is 85% efficient and its generator (kinetic→elec) is 94% efficient. Show that the turbine step is about 38% efficient. (3 marks)

Marking: set up the product (1), isolate the unknown by dividing (1), correct value matching the "show that" target (1). For a "show that", you must end at the stated number.

DECODE THE QUESTION

What the command word is really asking

◈ COMMAND WORDS

In calculation questions, the verb tells you exactly how much working to show. Match your response to it. Reach a numerical answer with units. Write each line of working — marks live in the method, not just the final number. Same as calculate — find the value, showing the steps that get you there. You're given the answer. Your job is the working that lands on it. Don't skip steps; finish on the stated value. State a figure for each source and an explicit point of difference (e.g. "hydro ~90% vs coal ~30% because hydro skips the thermal step"). Give the reason or mechanism — usually "energy is lost as heat at each conversion, and losses multiply." Link cause to effect. Name the form or value. No working needed — but be precise (say "thermal energy", not "energy").

Command word	What the marker wants
Identify / State	Name it — no explanation.
Describe	Give the features / what happens, in order.
Explain	Give the how / why — cause and effect.
Distinguish / Compare	State the difference (and similarity) explicitly.
Analyse	Break into parts and show how they relate.
Evaluate / Justify	Weigh strengths & limits, then a reasoned verdict.

Match your answer to the verb — the fastest marks in the exam.

DON'T LOSE EASY MARKS

Six traps that cost real marks

⚠ EXAM TRAPS

0.36 is a fraction; 36% is a percentage. Decide once whether you're working in decimals or percent and stay there. Fix: work in decimals, convert to % only at the very end. Putting total input on top gives an efficiency above 100% — impossible. Fix: useful output always goes on top; the answer must be ≲ 100%. The ratio only works if top and bottom share units. Fix: convert first — 1 kWh = 3.6 MJ — then divide. It contradicts the First Law and examiners penalise it. Fix: say energy is transferred to the surroundings as heat or degraded to a less useful form. Efficiency = how much of the input becomes electricity. Capacity factor = how much of the time a plant runs near full output. A wind farm can be ~40% efficient and have a ~35% capacity factor — different ideas. Fix: read which one the question wants.

WRITING TRAINER

The P·E·L move for "explain" parts

P · POINT answer the verb → **E · EVIDENCE** example / mechanism → **L · LINK** tie back to question

The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

✎ **P·E·L WRITING**

Calculation questions often end with a worded part — "explain why multi-step generation is less efficient." Examiners want a Point, Evidence, and a Link. Type an answer and watch the checklist light up. "A coal plant converts energy through four forms — chemical → thermal → kinetic → electrical — and energy is transferred to the surroundings as heat at each conversion. Because the step efficiencies multiply, the overall efficiency (~30%) is much lower than hydro. Hydro skips the chemical-to-thermal heat-engine step entirely, going potential → kinetic → mechanical → electrical, so it avoids the largest loss and reaches ~90%. Fewer steps and no heat engine mean hydro keeps far more of its input as useful electricity." Why it scores: names the forms, identifies the heat-engine step as the bottleneck, uses the "efficiencies multiply" rule, gives comparative figures, and links explicitly back to the comparison asked.

PRACTICE EXAM · 20 MARKS

Five VCAA-style questions

★ **PRACTICE & SELF-MARK**

Work each one on paper first. Then reveal the weak answer, the strong answer and the marking guide, and award yourself a score — it saves to your radial dock.

- 1 mark — correct ratio 320 ÷ 800 (useful over total).
- 1 mark — answer of 40% with the % sign.

- 1 mark — multiplies the three decimals (method).
- 1 mark — overall efficiency ≈ 32%.
- 1 mark — ≈ 646 MJ electricity (0.323 × 2000).

★ **PRACTICE & SELF-MARK**

"The student is wrong because the plant is only 30% efficient, so 70% is lost. That's just how power stations work." The energy is not destroyed. The First Law of Thermodynamics states energy cannot be created or destroyed, only transformed. In the coal plant, chemical energy becomes thermal, then kinetic, then electrical. The "wasted" 70% is real energy that has been transferred to the surroundings as low-grade thermal energy (heat) — in the cooling water, flue gases and friction — at each conversion step. It is degraded to a less useful form, not gone.

- 1 mark — energy is not destroyed / conserved.
- 1 mark — cites the First Law of Thermodynamics.
- 1 mark — names forms (chemical→thermal→kinetic→electrical).
- 1 mark — the waste is thermal energy transferred to the surroundings.

★ **PRACTICE & SELF-MARK**

"Hydro is better because it's renewable and cleaner. Coal is bad for the environment and only 30% efficient. Hydro is 90%." Hydro chain: gravitational potential → kinetic (falling water) → mechanical (turbine) → electrical. There is no thermal step, so it avoids the big heat-engine loss and reaches ~90%. Coal chain: chemical → thermal → kinetic → electrical. The chemical-to-thermal-to-kinetic stage is a heat engine, which loses a large fraction as waste heat (limited by the Second Law), pulling the overall to ~30%. Because step efficiencies multiply, coal's extra thermal step compounds the losses, so hydro keeps roughly three times as much of its input as useful electricity.

- 1 mark — correct hydro chain with named forms.
- 1 mark — correct coal chain with named forms.
- 1 mark — identifies the thermal / heat-engine step as the key loss.
- 1 mark — links to losses multiplying across steps.
- 1 mark — explicit comparative conclusion tied to the figures.

★ **PRACTICE & SELF-MARK**

"(a) Gas is about 90% because that's the biggest number. (b) Gas needs less energy. (c) Gas is better because it's more efficient." (c) By efficiency, gas wins — it needs far less input per unit of electricity. But efficiency alone is misleading: the solar input is free, renewable sunlight, whereas the gas input is a finite fossil fuel that releases CO_2. A low-efficiency renewable can still be the more sustainable choice.

- 1 mark — (a) multiplies decimals → ≈ 54%.
- 1 mark — (b) gas input ≈ 75 MJ (40 ÷ 0.536).
- 1 mark — (b) solar input ≈ 190 MJ (40 ÷ 0.21).
- 1 mark — (c) states which is more efficient with reason.
- 1 mark — (c) names a limitation of efficiency alone (e.g. free/renewable input, emissions).
- 1 mark — overall coherence: correct rearrangement of η = useful ÷ input throughout.

✦ **KEY TAKEAWAY**

Five things to walk into the exam knowing η = useful output ÷ total input × 100. Useful goes on top; the answer is always ≤ 100%. Multi-step efficiencies multiply (as decimals) — never add or average them. Wasted energy = input − useful, and it's transferred to the surroundings as thermal energy (heat) — never "lost" or "destroyed" (First Law). The thermal step is the bottleneck. Anything that burns fuel runs a heat engine, limited by the Second Law — that's why coal/gas sit low and hydro/wind sit high. Efficiency isn't everything — a low-efficiency renewable with a free, clean input can still be the more sustainable choice.

EXAM TECHNIQUE

★ Worked answers — weak vs strong

The same question answered two ways. The strong column shows the moves that earn the marks.

P · POINT answer the verb → **E · EVIDENCE** example / mechanism → **L · LINK** tie back to question

The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

800 − 320 = 480, so it's 480 kJ efficient." Confuses the wasted energy with efficiency, gives no ratio, no percentage. This is a wasted-energy calculation, not an efficiency one.

✓ STRONG / MODEL ANSWER

η = useful output ÷ total input × 100 = 320 ÷ 800 × 100 = 40%. Correct formula, substitution shown, units cancel, percentage stated.

✗ WEAK ANSWER

85 + 40 + 95 = 220, divided by 3 = 73% efficient. So 0.73 × 2000 = 1460 MJ." Averaging the percentages is the classic multi-step error — efficiency can't be found by adding or averaging. Both the method and both numbers are wrong.

✓ STRONG / MODEL ANSWER

η overall = 0.85 × 0.40 × 0.95 = 0.323 = 32% (≈ 32.3%). Electricity out = 0.323 × 2000 = ≈ 646 MJ. Multiplies decimals, converts to %, then applies it to the input.

✗ WEAK ANSWER

The student is wrong because the plant is only 30% efficient, so 70% is lost. That's just how power stations work." Repeats the misconception ("lost"), names no forms, cites no law, and doesn't correct the "destroyed" claim. Only the efficiency figure earns anything.

✓ STRONG / MODEL ANSWER

The energy is not destroyed. The First Law of Thermodynamics states energy cannot be created or destroyed, only transformed. In the coal plant, chemical energy becomes thermal, then kinetic, then electrical. The "wasted" 70% is real energy that has been transferred to the surroundings as low-grade thermal energy (heat) — in the cooling water, flue gases and friction — at each conversion step. It is degraded to a less useful form, not gone. Corrects the error, names forms, cites the First Law, explains where the energy actually goes.

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Worked good/weak answers illustrate exam technique — always check against current VCAA marking advice.